



May 2025

Max. Marks: 100

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

- 1-

12. The expansion device in any refrigeration system is connected between
 A) Evaporator and Condensor B) Evaporator and Compressor
 C) Compressor and Condensor D) Evaporator and Absorber
13. Worm gear is used, when two shafts are at
 A) Right angles to each other B) Some distance apart
 C) Radial direction D) 45 cm apart
14. To obtain a higher torque, driven gear size should be,
 A) Smaller than driver B) Larger than driver
 C) Same as driver D) Do not depend on gear size
15. Which of the following is a type of fusion welding
 A) Ultrasonic welding B) Carbon arc welding
 C) Forge welding D) Resistance welding
16. The main purpose of flux in soldering is to
 A) Keep the metal surface clean and oxide free B) Reduce melting point
 C) Ensure proper texture D) Reduce temperature
17. A lathe operation in which the bar length is reduced
 A) Facing B) Knurling
 C) Plain turning D) Tapering
18. Which of the following operation does NOT use a rotating tool?
 A) Reaming B) Counter-Boring
 C) Facing D) End milling
19. Fixed automation system is suitable for a production line in which,
 A) Only one variety of product produced B) Multiple variety of products produced
 C) Limited varieties of product produced D) Product varieties slightly differ from each other
20. Cartesian configuration of robot anatomy has,
 A) Only spherical movements B) Both circular and linear
 C) Only linear movements D) Both special and linear

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Define the following with examples: i) Conventional energy sources ii) Renewable energy sources.	4	L*1	1	1
b)	A steam at 10 bar and dryness fraction 0.98 receives 140 kJ/kg at the same pressure. What is the final state of the steam? At 10 bar the following values are noted, $T_s = 179.9^\circ\text{C}$ $h_f = 762.61 \text{ kJ/kg}$ $h_{fg} = 2013.6 \text{ kJ/kg}$ Specific heat for superheated steam = 2.25 kJ/kgK	4	L2	1	1
c)	Explain the working of centrifugal pump with neat sketch.	8	L2	2	1
2. a)	Explain the working principle of Babcock & Wilcox boiler with neat sketch.	8	L2	1	1
b)	Differentiate between reaction and impulse steam turbine.	8	L2	2	1
3. a)	Give the functions of the following used in the steam generator: i) Boiler ii) Steam stop valve iii) Fusible Plug iv) Economizer.	8	L2	1	1
b)	Explain the working of single stage reciprocating compressor with neat sketch.	8	L2	2	1
Unit – II					
4. a)	A single cylinder four stroke engine runs at 1000 rpm and has a bore of 115 mm and has a stroke of 140 mm. The brake load is 6 kg at 600 mm radius and the mechanical efficiency is 80%. Calculate brake power and mean effective pressure.	8	L2	3	1
b)	Describe the principle of arc welding with suitable welding circuit diagram.	8	L2	4	1

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|----|----|---|---|----|---|---|
| 5. | a) | With a neat sketch, explain the working of vapor compression refrigeration. | 8 | L2 | 3 | 1 |
| | b) | Mention any four advantages of gear drive. | 4 | L2 | 4 | 1 |
| | c) | Define soldering and brazing techniques. | 4 | L2 | 4 | 1 |
| 6. | a) | Explain the working of four stroke petrol engine with neat sketch. | 8 | L2 | 3 | 1 |
| | b) | In a crossed belt drive, the difference in tensions between the tight and slack sides of the belt is 1000 N. Find the tensions on the slack and tight sides, if the angle of contact is 160° and the coefficient of friction is 0.3. | 8 | L2 | 4 | 1 |

Unit – III

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|----|-----|--|---|----|---|---|
| 7. | a) | Explain the following lathe operations: | | | | |
| | i) | Facing | | | | |
| | ii) | Thread Cutting. | 8 | L2 | 5 | 1 |
| | b) | Explain the following types of automation: | | | | |
| | i) | Fixed automation | | | | |
| | ii) | Flexible automation. | 8 | L2 | 5 | 1 |
| 8. | a) | Define a robot and explain the classification of robot based on physical configuration with neat sketches. | 8 | L2 | 5 | 1 |
| | b) | Explain the following with neat sketch: | | | | |
| | i) | Up milling | | | | |
| | ii) | Down milling. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
